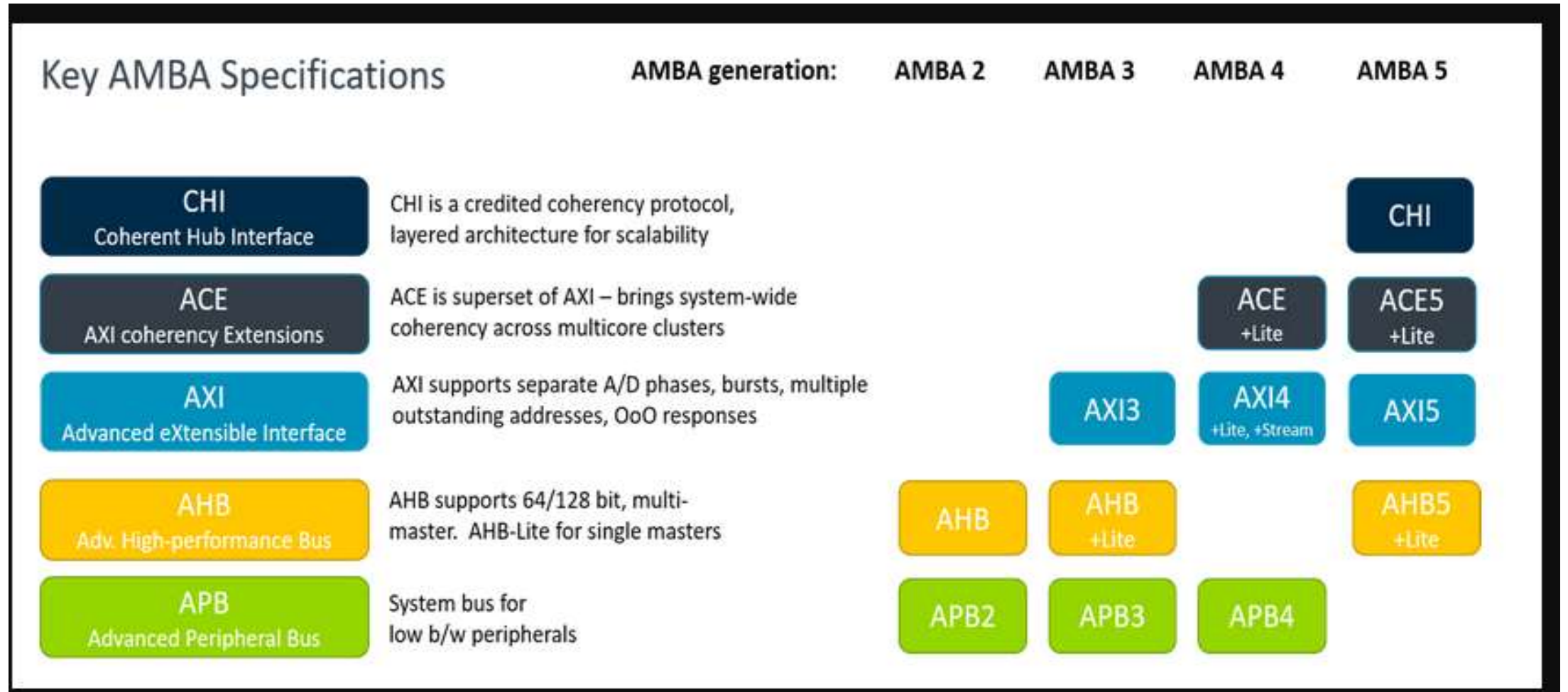


CHI (Coherent Hub Interface)

Introduction

- Coherent Hub Interface (CHI) is an evolution of the AXI Coherency Extensions (ACE) protocol. It is part of the Advanced Microcontroller Bus Architecture (AMBA).
- AMBA is a freely available, globally adopted, open standard for the connection and management of functional blocks in a system-on-chip (SoC).
- It facilitates right-first-time development of multi-processor designs, with large numbers of controllers and peripherals.

AMBA ROADMAP



ACE vs CHI

- ACE was designed as an extension to AXI to handle coherency, but it is not without shortfalls. It served designs with smaller coherent clusters well, but as SoCs and system becomes more complex and the number of processors increased, the need for better coherency and efficiency increased.
- Arm released a packet-based layered coherency architecture, without dependencies on AXI or ACE. CHI was built with performance improvement and scalability in mind of applications and systems such as mobile, networking, automotive and data center.

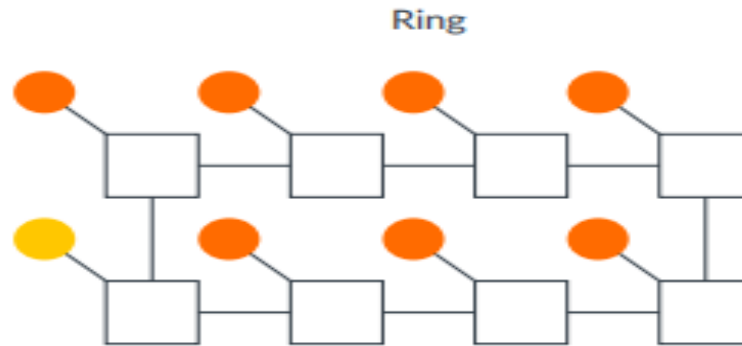
AMBA 4 ACE	AMBA 5 CHI
Dedicated channels for different functions: write (AW, W, B), read (AR, R), snoop (AC, CR, CD)	Generic signals for all functions (TX, RX) with transaction type encoded in the data transfer
Valid/Ready Flow control makes is easy to observe back pressure	Flow control is credit based and more difficult to observe
Easy to understand address map routes transactions to ports	Address map routing is more complex
Internal details of interconnect not required to understand functionality	Internal functionality of interconnect plays a larger role in understanding system behavior

Components of Interconnect

- Standalone processors
- Processor clusters
- Graphic processors
- Memory controllers
- I/O bridges
- PCIe subsystems
- Interconnects

RING TOPOLOGY

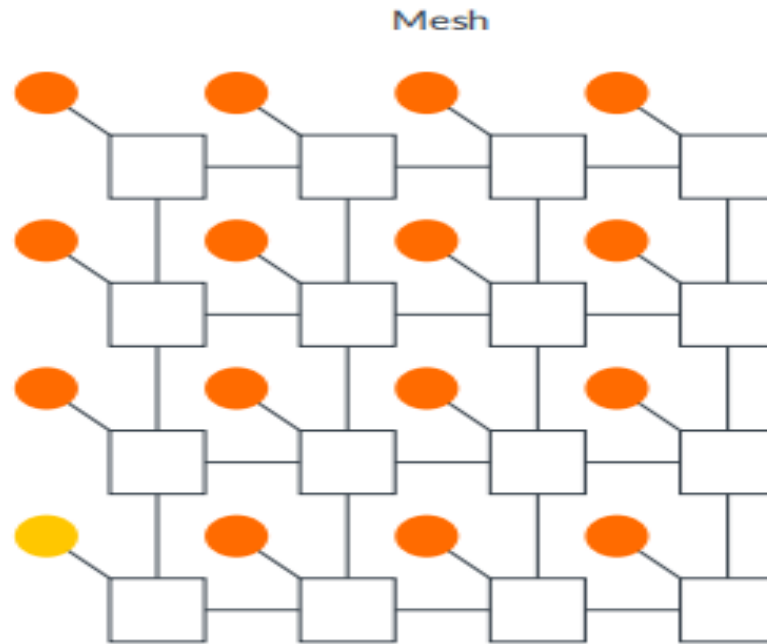
- In the ring, each component connects directly to two other components, forming a ring where all the components can communicate with each other.
- The disadvantage of this topology is that latency increases linearly with the number of components in the ring. This is because a transaction must traverse the ring until it reaches its destination.
- The ring topology is best suited for medium-sized systems.



In this diagram, the circles represent requester and subordinate components in the network. The squares represent intermediary components to route transactions between requester and subordinate.

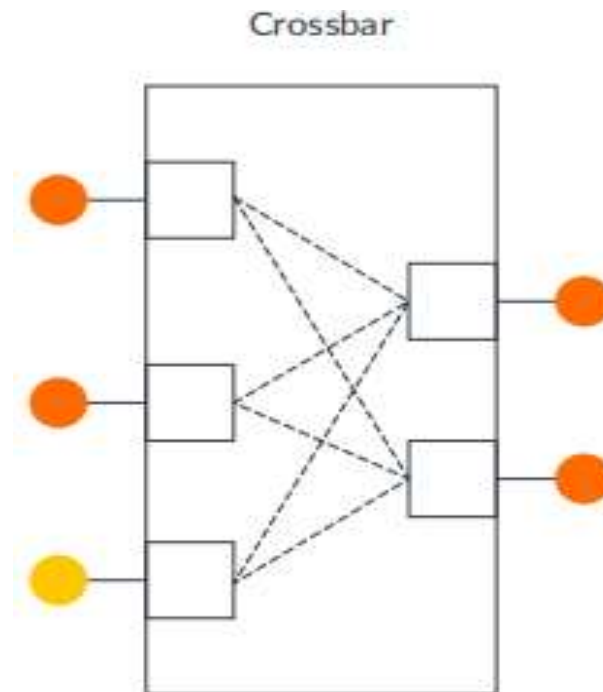
MESH TOPOLOGY

- Compared to the ring, the mesh contains more paths for a transaction to reach its destination and therefore reduces the travel time of a transaction.
- This provides higher bandwidth in the system, at the cost of more area.
- The mesh topology is best suited for large scale systems.



CROSSBAR TOPOLOGY

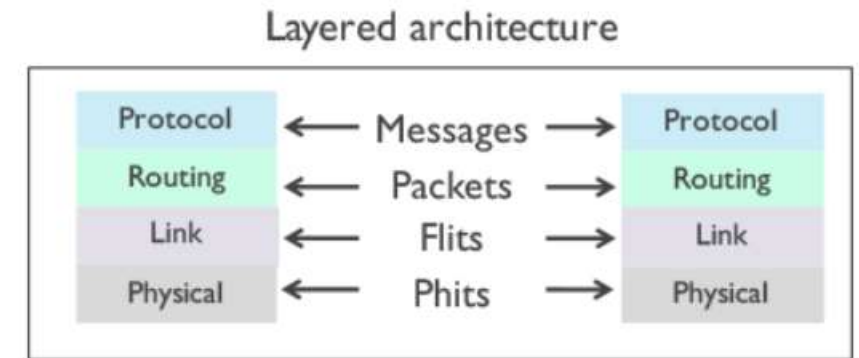
- This topology allows every node to connect to every possible node.
- The drawback of this topology is the cost of connecting all the components. This is because the number of wires needed in the system can grow significantly with each additional component.
- The crossbar topology is best suited for small-sized systems.



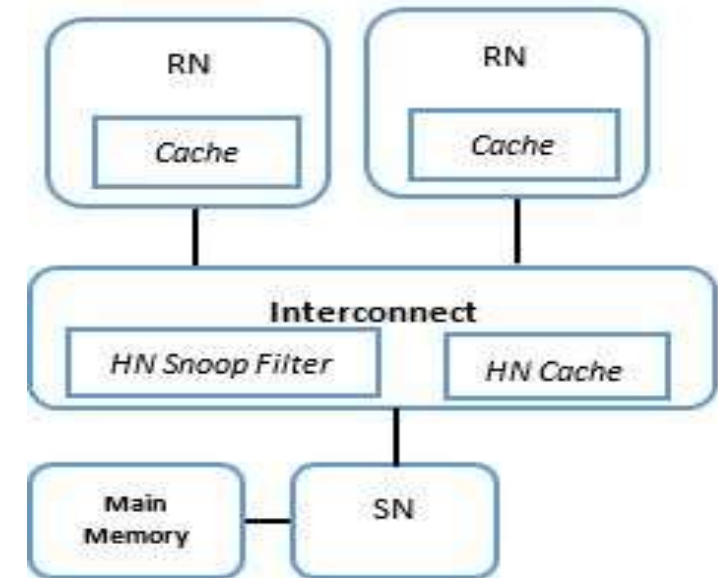
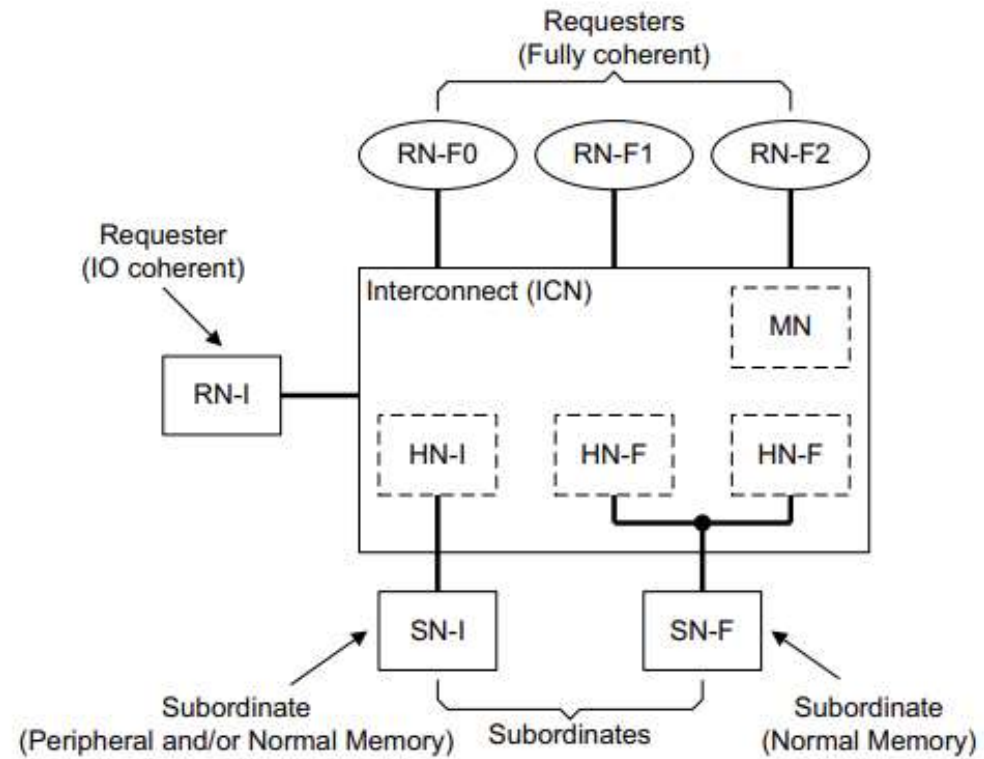
Architecture Layers

Functionality is grouped into the following layers:

- The **Protocol layer** is the topmost layer in the CHI architecture. The function of the Protocol layer is to :
 - Generate and process requests and responses at the protocol nodes.
 - Define the transaction flows for each request type.
 - Manage the protocol level flow control.
- The function of the **Network layer** is to :
 - Packetize the protocol message.
 - Determine the source and target Node IDs required to route the packet over the interconnect to the required destination and add to the packet
- The function of the **Link layer** is to :
 - Provide flow control between network devices.
 - Manage link channels to provide deadlock-free switching across the network.



Example CHI system



Component Naming

RN :

Request Node generates protocol transactions, including reads and writes, to the interconnect.

RN-F :

Fully Coherent Request Node

- Includes a hardware-coherent cache.
- Permitted to do all transactions as defined by the protocol.
- Supports all Snoop transactions.

RN-D :

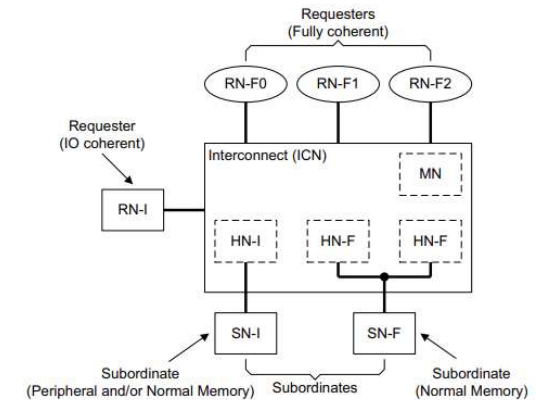
IO Coherent Request Node with DVM support

- Does not include a hardware-coherent cache
- Receives DVM transactions.
- Generates a Subset of transactions defined by the protocol.

RN-I :

IO Coherent Request Node

- Does not include a hardware-coherent cache
- Does not receives DVM transactions.
- Generates a Subset of transactions defined by the protocol.
- Does not require Snoop functionality.



Component Naming(Contd..)

HN :

Home Node within the interconnect receives protocol transactions from Request Nodes.

HN-F :

Fully Coherent Home Node

- Expected to receive all request types, except DVMOp.
- Includes a PoC (Point of Coherence) and PoS (Point Of Serialization).
- Might include a directory or snoop filter to reduce redundant snoops

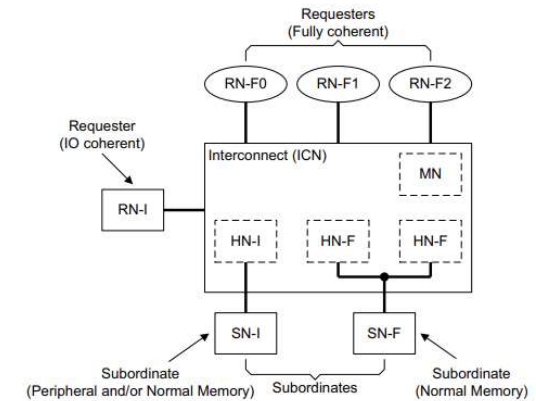
HN-I :

Non Coherent Home Node

- Processes a limited subset of transactions defined by the protocol.
- Does not includes PoC and is not capable of processing a snoopable request.
- Expected to be the PoS that manages order between IO requests targeting the IO subsystem.

MN :

Miscellaneous Node receives a DVM transaction from a Request Node, completes the required action and returns a response.



Component Naming(Contd..)

SN :

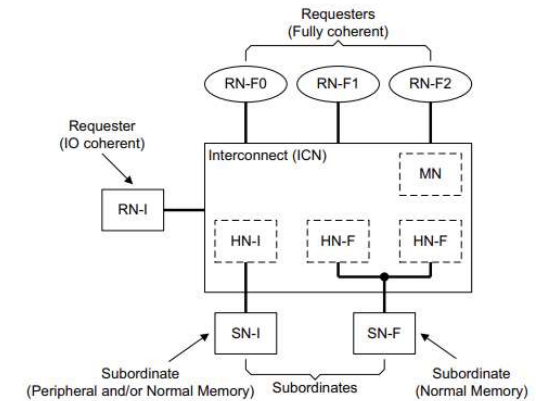
Subordinate Node receives a request from Home Node, completes required action, and returns a response.

SN-F :

A Subordinate Node type used for Normal memory. It can process Non-snoopable Read, Write, and Atomic requests, including exclusive variants of them, and Cache Maintenance Operation (CMO) requests.

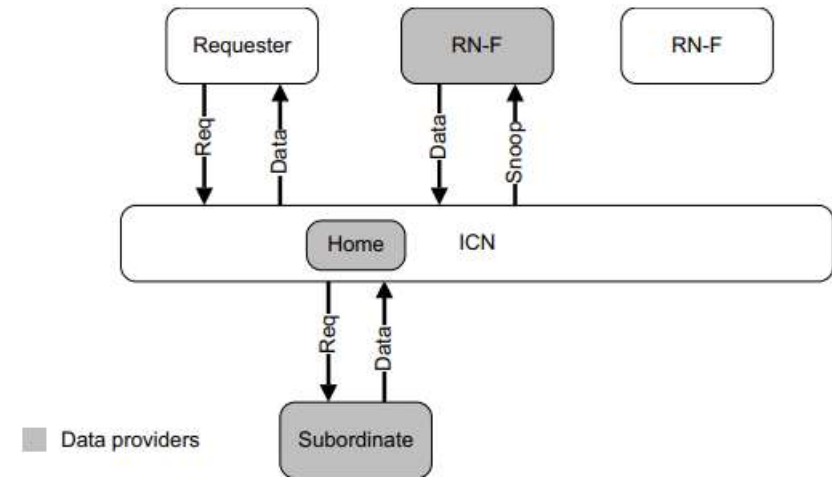
SN-I :

A Subordinate Node type used for Peripherals or Normal memory. It can process Non-snoopable Read, Write, and Atomic requests, including exclusive variants of them, and Cache Maintenance Operation (CMO) requests.

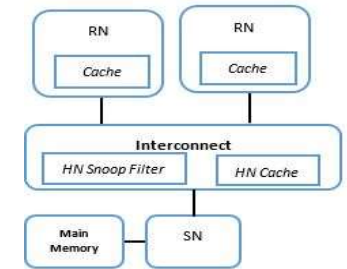
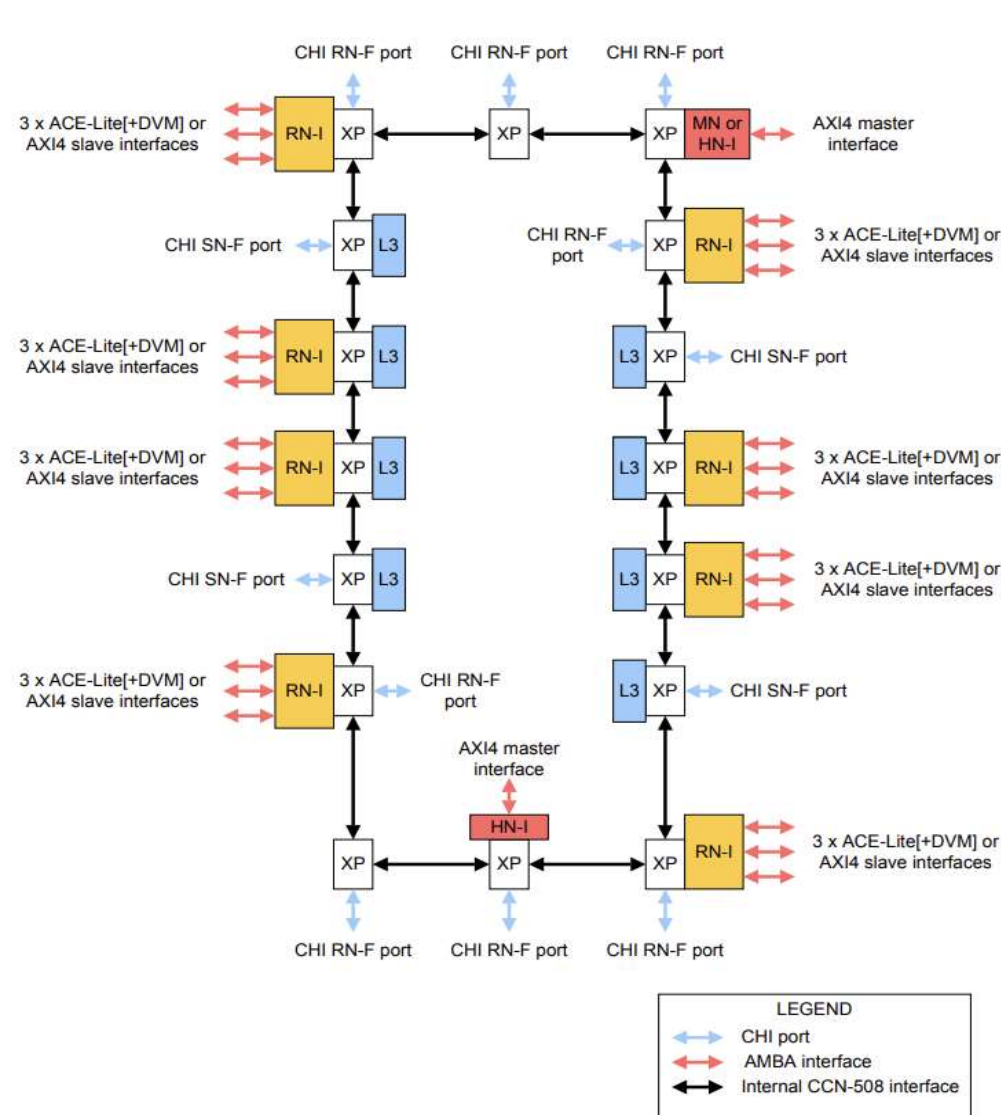


Read Data Source

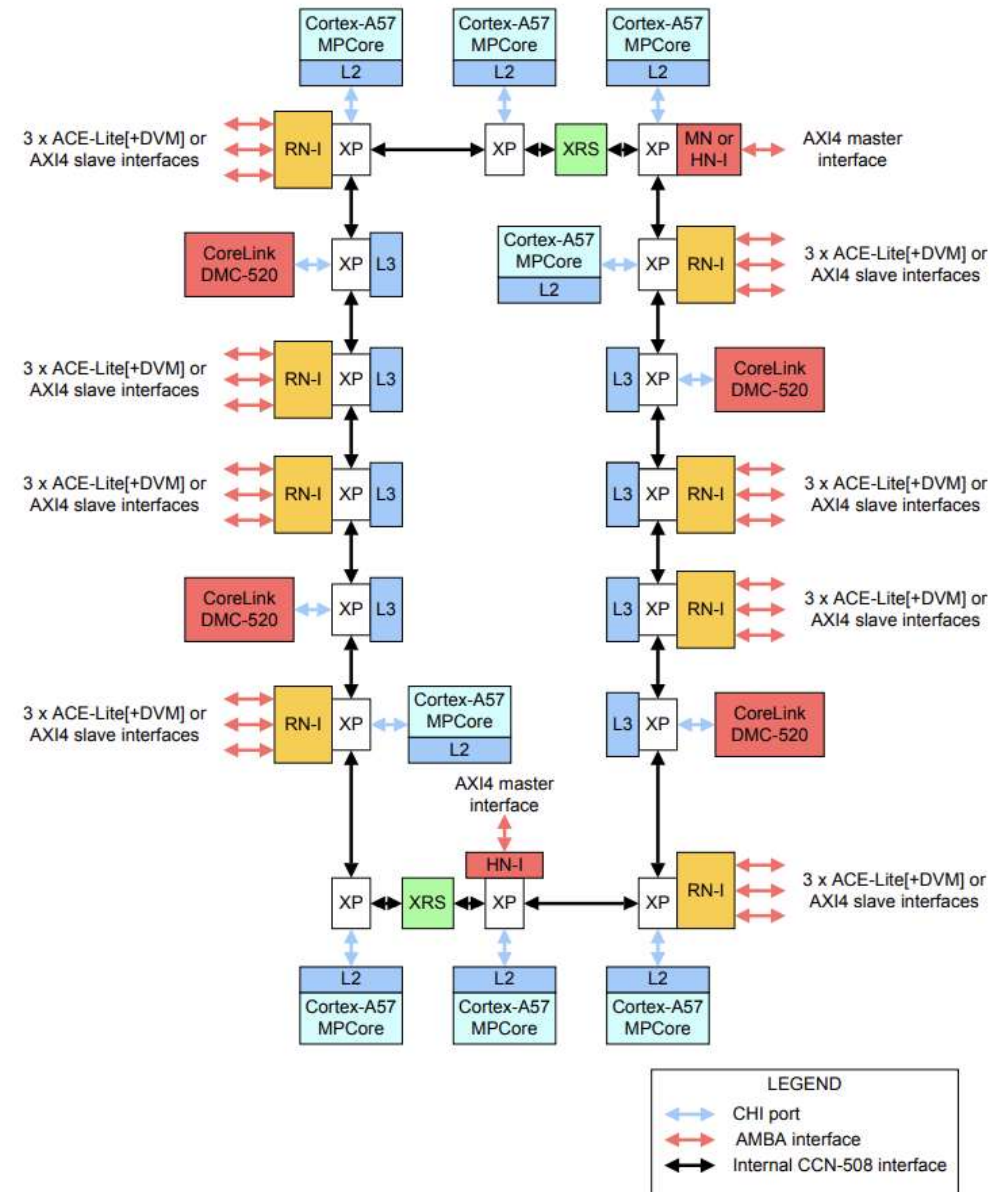
- The data for a read request can be either from the Cache within Home or Subordinate or from Peer RN-F
- CHI uses three features to reduce number of hops to complete a transaction.
 - ☐ **DMT** Direct Memory Transfer
 - ☐ **DCT** Direct Cache Transfer
 - ☐ **DWT** Direct Write-data Transfer

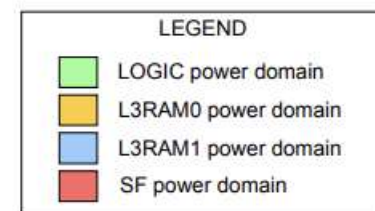
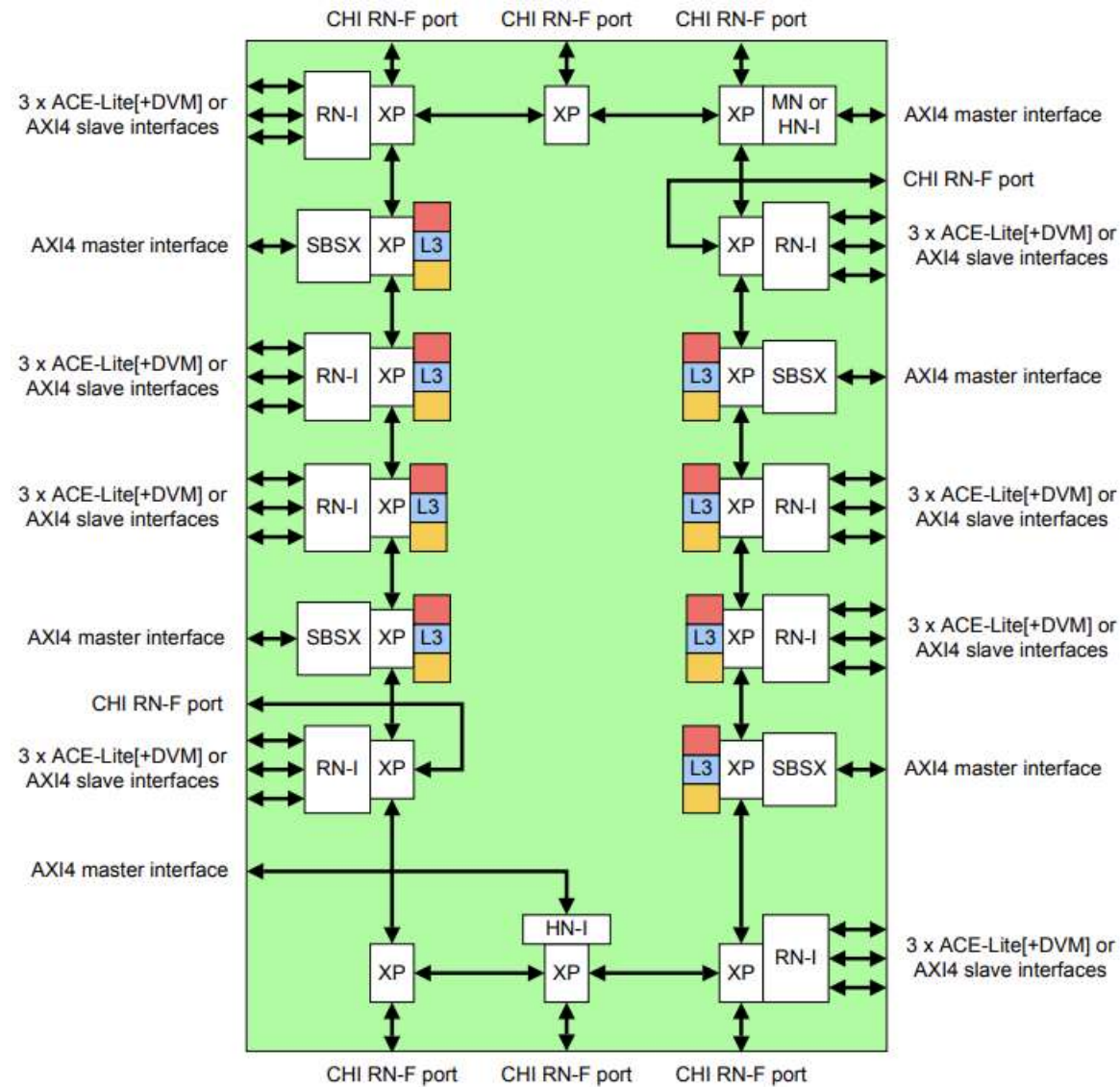


ARM® CoreLink™ CCN-508 Cache Coherent Network



ARM® CoreLink™ CCN-508 Cache Coherent Network with complete coherent system.



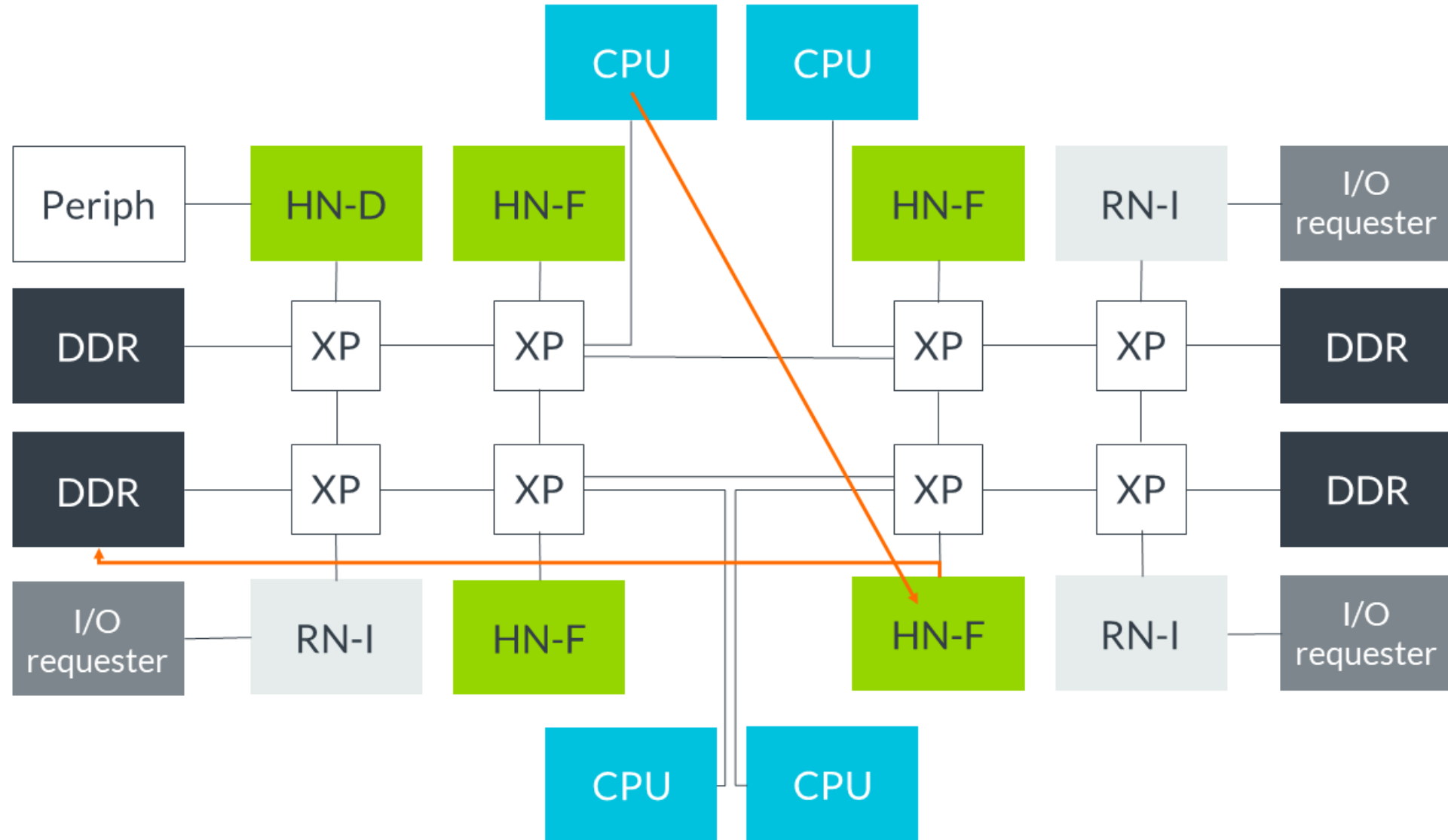


CCN details

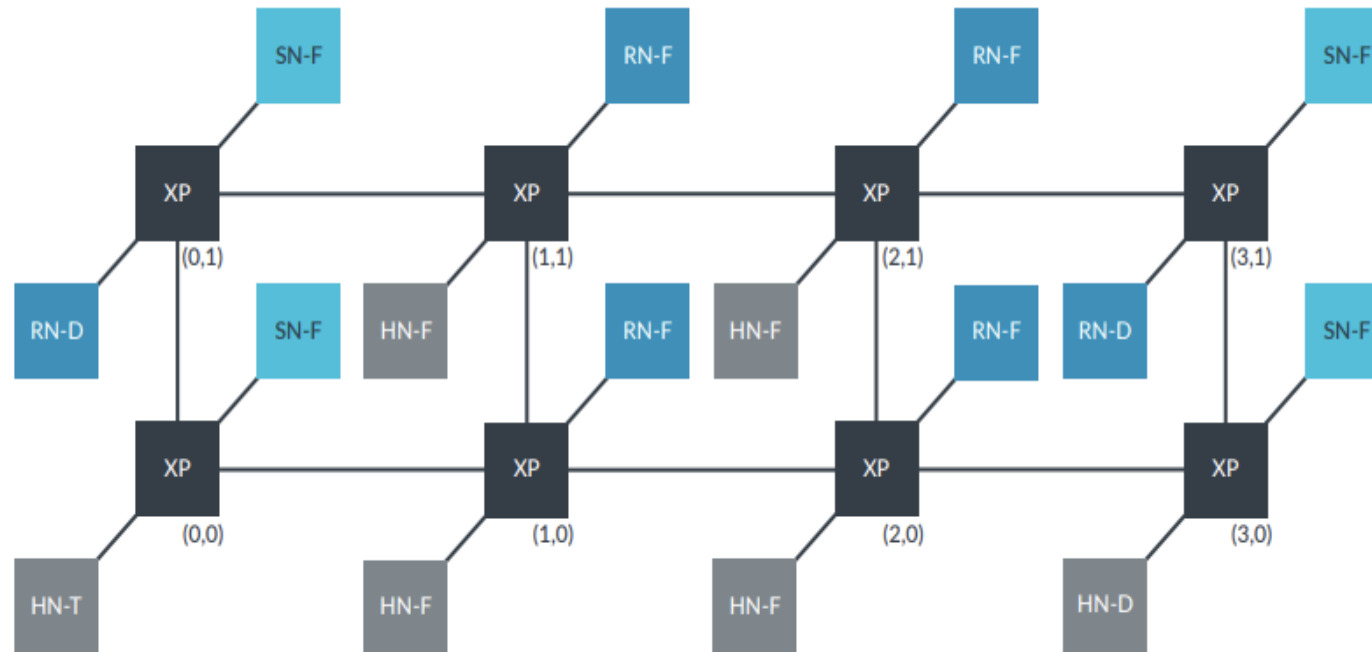
- XP : CrossPoint is switch or router logic module
- RN-F : Processors, clusters, GPUs or other RNs with a coherent cache.
- SN-F : CHI memory controllers
- HN-F : Have SLC and SF, have combined PoC and PoS which is responsible for ordering of all memory requests sent to this HN-F.
- SLC : System Level Cache(L3 here), allocation policy is “exclusive until shared”.
- SF : Snoop Filter for tracking cache line states in RN-Fs.

In a CHI interconnect, you assign a single HN to each byte of the system address space. That HN is responsible for handling all memory transactions that are associated with that address.

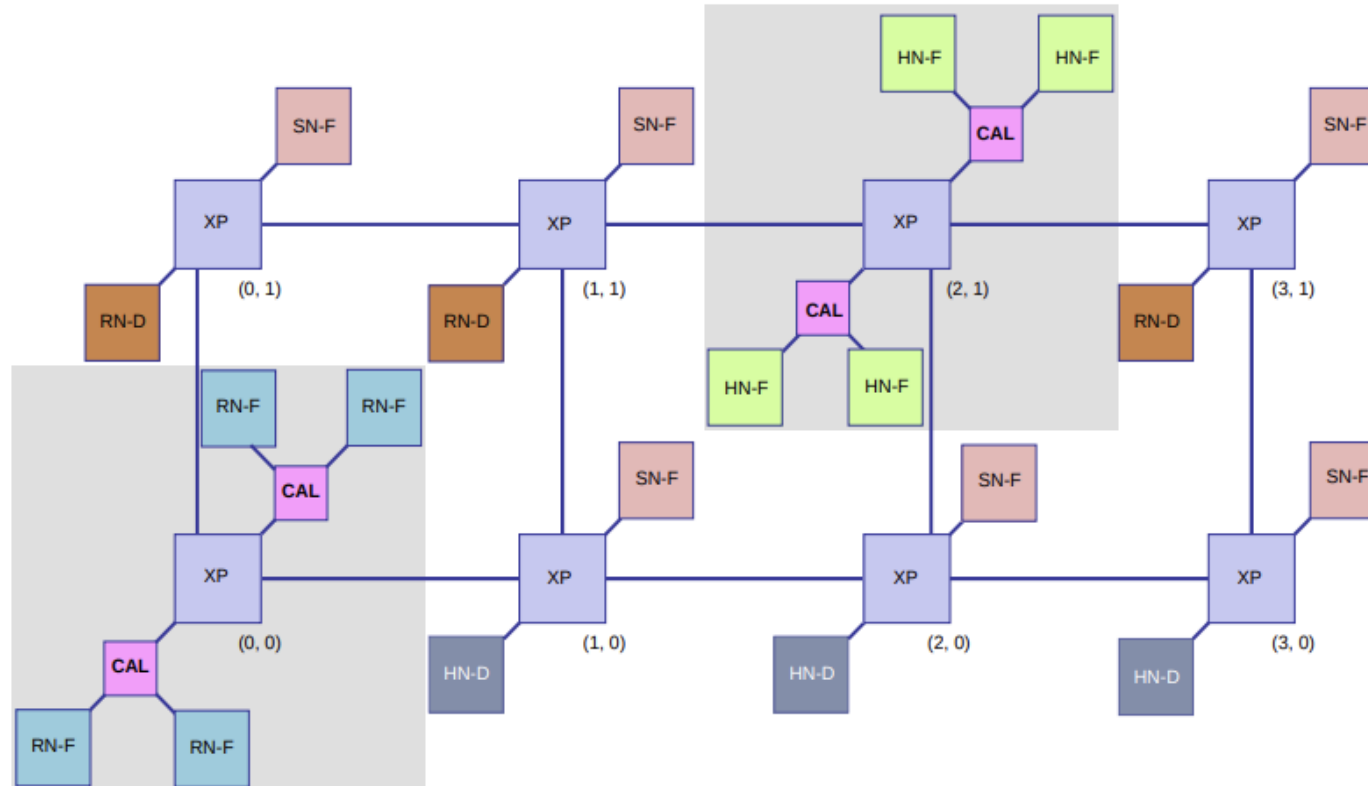
RN -> HN -> SN



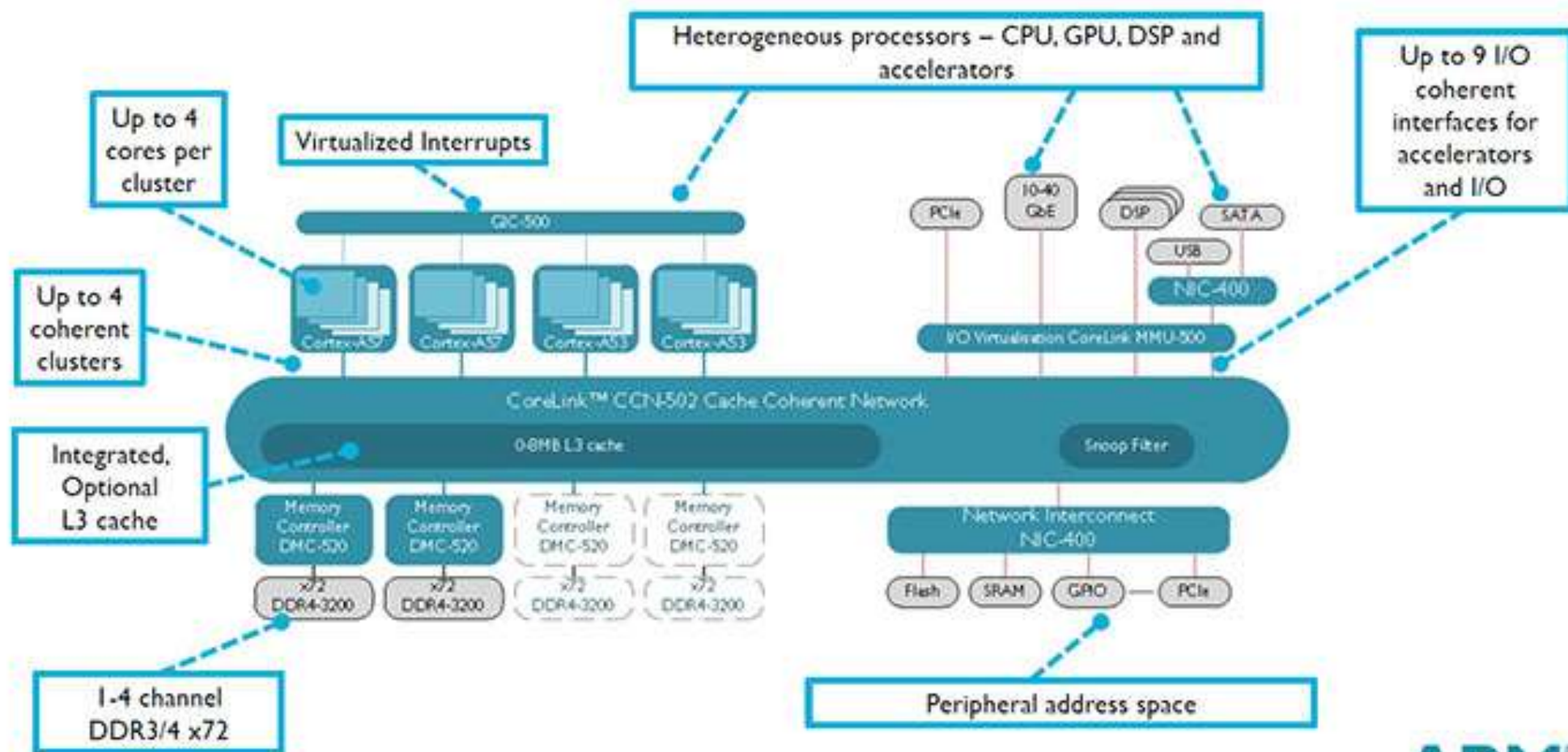
Arm® CoreLink™ CI-700 Coherent Interconnect 4 x 2 mesh configuration



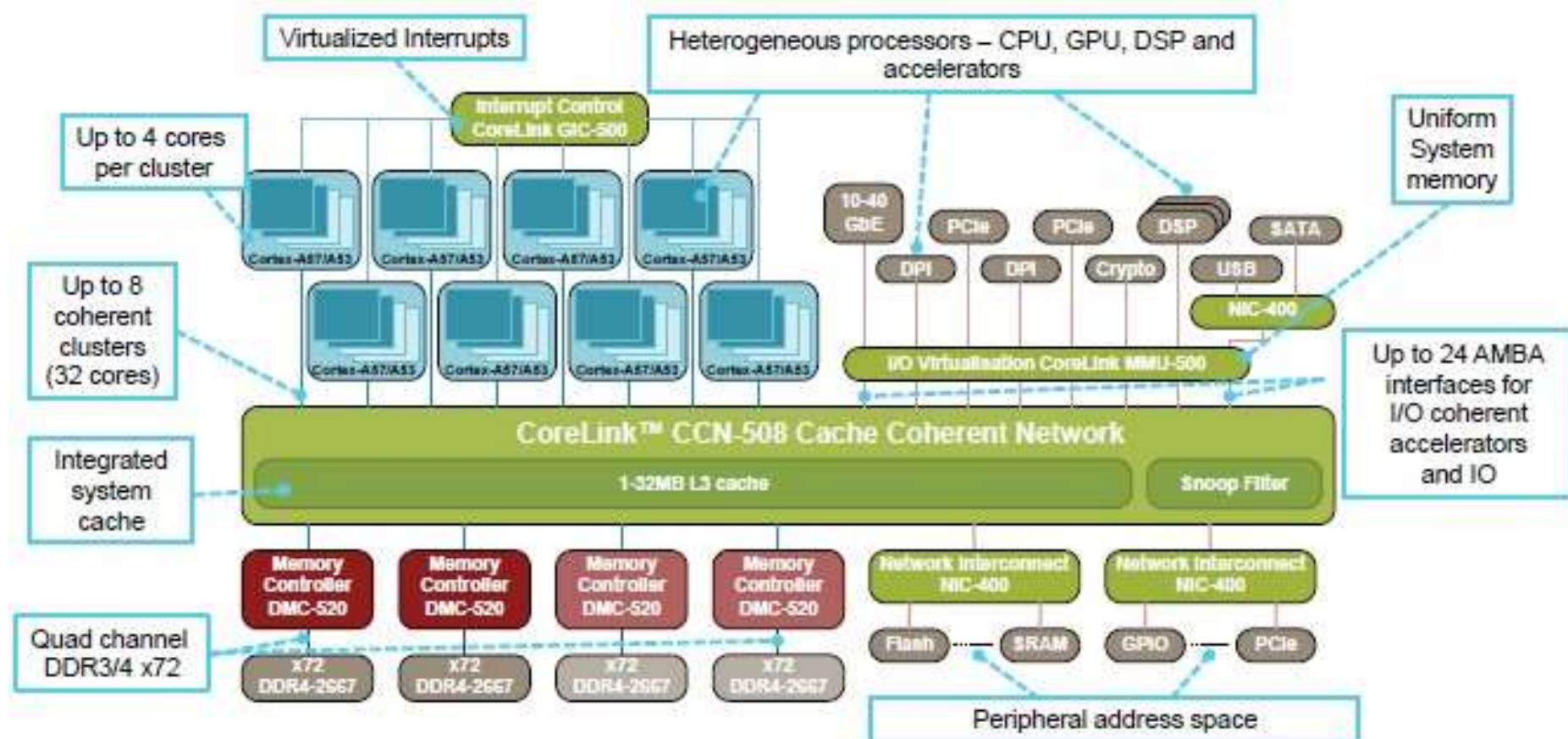
Arm® CoreLink™ CI-700 Coherent Interconnect 4 x 2 mesh configuration with CAL



ARM's CCN-502 - High Performance, Optimized Footprint

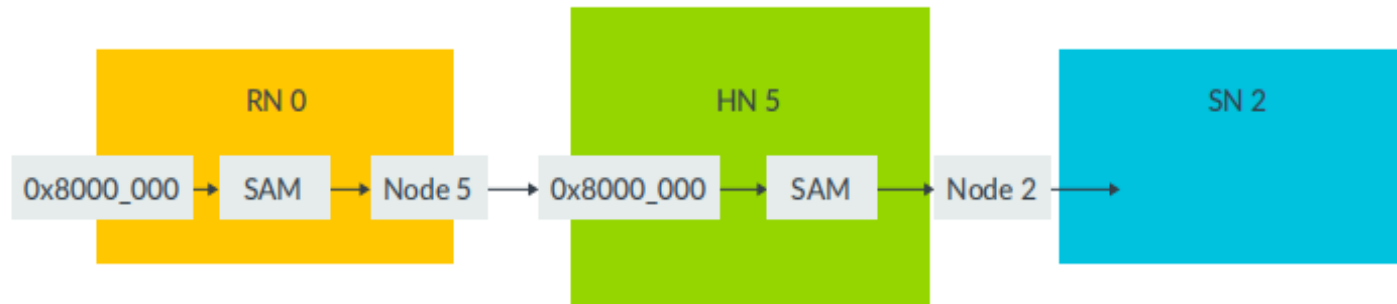


ARM®



System Address Map

- Every component in the system is assigned a Unique Node ID. CHI uses the System Address Map (SAM) to convert physical addresses to a target Node ID.
- To be able to determine the target Node ID of outgoing requests, each RN and HN must have a SAM.

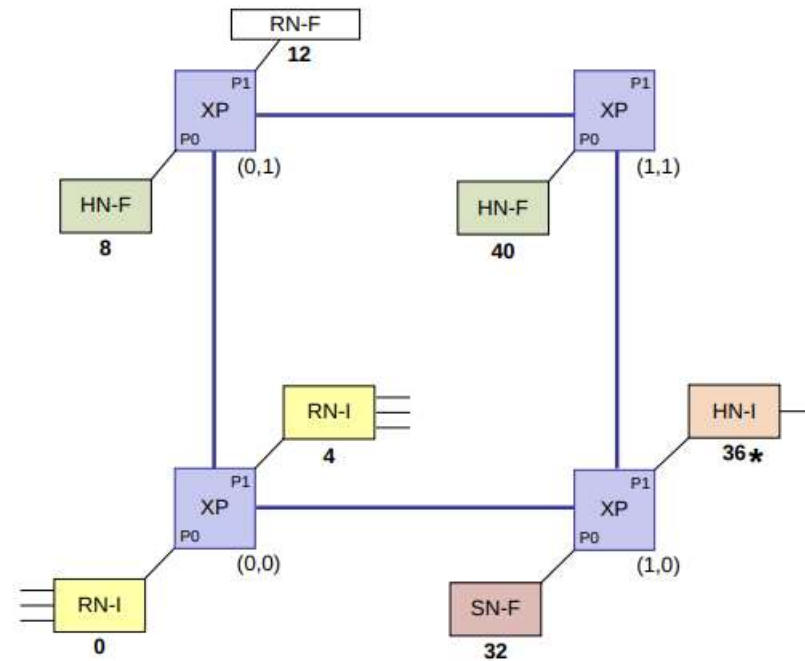


System Address Map(Contd...)

- 1.The transaction with address 0x8000_0000 passes through the RN SAM in Node 0.
2. The RN SAM determines the destination as Node 5.
3. The transaction is routed to the HN with Node 5.
4. The HN receives the transaction.
5. The HN passes the address through its HN SAM and determines the destination as Node 2.
6. The transaction gets routed to the SN with Node 2.

NODE IDs

- For the HN-I connected to XP (1,0), the node ID reads as 36.
- The equivalent binary value is 01 00 1 00.
- In other words, the X position value = 01, the Y position value = 00, the device port value = 1, and the device ID value = 00.



CHANNELS

RN channel designation

TXREQ. Outbound Request.

TXDAT. Outbound Data.

Used for write data, atomic data, snoop data, forward data.

TXRSP. Outbound Response.

Used for snoop response and completion acknowledge.

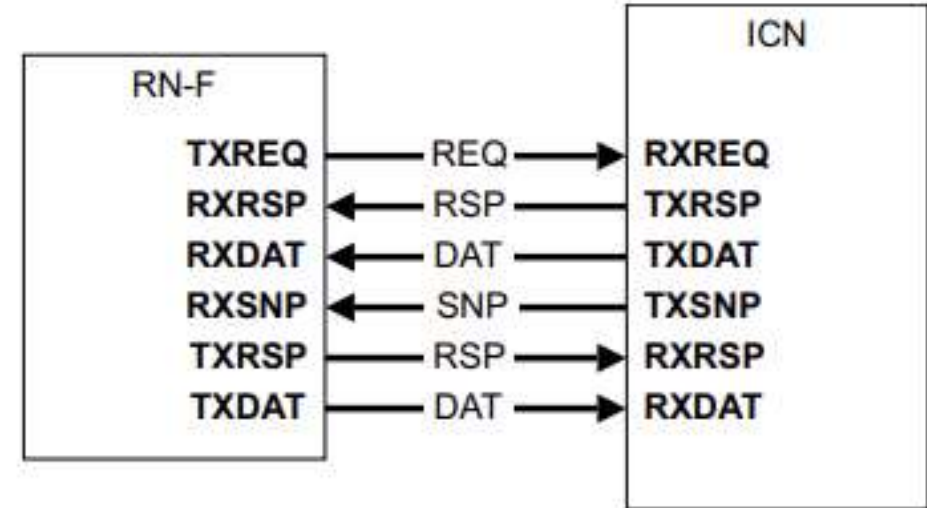
RXRSP. Inbound Response.

Used for responses from the Completer.

RXDAT. Inbound Data.

Used for read data, atomic data.

RXSNP. Inbound Snoop request.



Key Features

- Scalable architecture
- Independent layered approach, comprising of Protocol, Network, and Link layer, with distinct functionalities.
- Packet-based communication.
- All transactions handled by an interconnect-based Home Node that co-ordinates required snoops, cache, and memory accesses.
- The CHI coherence protocol supports:
 - Coherency granule of 64-byte cache line.
 - Snoop filter and directory based systems for snoop scaling.
 - Both MESI and MOESI cache models with forwarding of data from any cache state.
 - Additional partial and empty cache line states.
- Supports Virtual Memory Management through DVM operations.
- Supports Cache Stashing and atomic operations
- Request Retry to manage protocol resources.
- Support for end-to-end QoS.
- RAS features.
- Support for MTE(Memory Tagging Extension).

THANK YOU